COMP7610 Introduction to Web3 and Blockchain – Assignment 3

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1. On-chain club system deployment

The HKU Drone Club system contains two smart contracts. `ClubPointToken` is an ERC-20 token named Club Point Token (symbol `CPT`) with zero decimals, so all amounts are whole CPT. `CampusMemoNFT` is an ERC-721 collection named Campus-Memo (symbol `CAMPUS`) with a maximum supply of 10.

Deployment steps in Remix:

1. Compile `scripts/ClubPointToken.sol` and deploy `ClubPointToken` using the selected deployer account. Record the address: [0x1e26F4F549ff92eE6C2760de4B0b236A73979DF5].
2. Compile `scripts/CampusMemoNFT.sol`. Deploy `CampusMemoNFT` with the CPT address from Step 1 and the metadata base URI:

```
{
  "name": "Campus-Memo",
  "description": "COMP7610 assignment 3 NFT URI",
  "image": "https://github.com/phpw0929/COMP7610/blob/main/assignment%203/assets/drone_icon.
  ↪ png?raw=true"
}
```

   Then record the address: [0x64332ac79b2617a6B5ab2071dd868032dF83214b].
3. In the deployed CPT contract, call `setCampusMemoContract` from the `CampusMemoNFT` contract. This one-time owner-only configuration authorizes the contract to burn CPT for NFT purchases.
4. Final deployed addresses:
   - ClubPointToken = [0x1e26F4F549ff92eE6C2760de4B0b236A73979DF5];
   - CampusMemoNFT = [0x64332ac79b2617a6B5ab2071dd868032dF83214b];
   - NFT metadata URI = [https://gist.githubusercontent.com/phpw0929/24a5edbe6d2a7d4d706b2d6efd3c0546/raw/a7e25fe8baa711d375008dac9c6c17b37a1ffcb3/Campus-Memo_URL.json].

2. Custom function and additional logic

Custom function – `applyForClubWithdrawal()`. An active member can withdraw immediately. The function burns the caller's entire CPT balance, sets `activeMember` to false, sets `hasWithdrawn` to true, and emits `ClubWithdrawal` event. A withdrawn member cannot register again or receive rewards.

Additional functional logic. The system also provides:

- Member registration through `registerMember()`;
- Whole-number CPT transfers using the inherited ERC-20 `transfer()`;
- Owner-only single and batch CPT rewards through `mintCPT()` and `distributeCPT()`;
- Campus-Memo minting. Each NFT costs 100 CPT, which is burned directly, and token IDs run from 1 to 10. NFT transfers use standard ERC-721 behavior. Events record registration, rewards, withdrawal, contract linking, and NFT minting.

3. Function testing summary

All tests were performed in Remix after deploying both contracts. CPT amounts below are displayed directly as whole CPT because `decimals()` returns 0.

Test	Action and observed result
Initial state	**Passed**. Confirmed name, symbol, zero decimals, 1,000 CPT total supply, 1,000 CPT assigned to the deployer, and deployer active status. .
Contract linking	**Passed**. Owner successfully called `setCampusMemoContract`, which links the two contracts; a second call is reverted because the address can be set only once.
Registration	**Passed**. A new account successfully called `registerMember`; repeated registration is reverted.
Owner minting	**Passed**. Owner minted 200 CPT to a member; balance and total supply increased correctly. A non-owner mint attempt is reverted.
CPT transfer	**Passed**. A member transferred 50 CPT. Sender and recipient balances changed by 50; transfers used integer CPT values. An over-balance transfer is reverted.
Batch rewards	**Passed**. Owner distributed 10 CPT and 30 CPT to two active members in one batch. The operation is reverted when the CPT value array-length mismatch with the member array-length.
NFT minting	**Passed**. An eligible member minted a Campus-Memo with token ID 1. Exactly 100 CPT was burned; `ownerOf(1)` returned the minter and `tokenURI(1)` returned the expected metadata URI.
NFT restrictions	**Passed**. An account with less than 100 CPT could not mint. After token IDs 1–10 were minted, an additional mint is reverted with error message 'supply exhausted'.
Withdrawal	**Passed**. An active member withdrew; the entire CPT balance became zero, membership became inactive, and `hasWithdrawn` became true. Re-registration and further NFT minting is reverted.
NFT transfer	**Passed**. The NFT owner transferred the Campus-Memo of token 1 to another account. As expected, `ownerOf(1)` changed while CPT balances were unchanged.

4. Submission details

Student name and UID:	WONG Pak Hung Philip (3036849206)
ClubPointToken address:	0x1e26F4F549ff92eE6C2760de4B0b236A73979DF5
CampusMemoNFT address:	0x64332ac79b2617a6B5ab2071dd868032dF83214b
NFT base/token URI:	https://gist.githubusercontent.com/phpw0929/24a5edbe6d2a7d4d706b2d6efd3c0546/raw/
Network	Sepolia testnet
Deployment date	11/10/2026

The source code is submitted separately as `ClubPointToken.sol` and `CampusMemoNFT.sol`. Replace every red placeholder and rename this file to the required full-name/UID filename before submission.